

Finitely generated cones are polyhedral cones

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In this note we use the projection theorem for polyhedral cones to argue that finitely generated cones are also polyhedral cones. The argument is short. However, the result is essential for obtaining the remaining results on polyhedra. A complete note is therefore dedicated to establish this fact.

The projection theorem for polyhedral cones states that the projection of a polyhedral cone is also a polyhedral cone (proved in note 1.(C)):

Theorem 1 (*The projection theorem for polyhedral cones*)

For any polyhedral cone $C \subseteq \mathbb{R}^{n+k}$, the set obtained by projecting out the last k variables:

$$Proj_k(C) = \{x \in \mathbb{R}^n : \exists y \in \mathbb{R}^k : \begin{pmatrix} x \\ y \end{pmatrix} \in C\}$$

is a polyhedral cone. ■

The argument is now as follows. Let $\text{cone}(E)$ be a finitely generated set, where $E \subset \mathbb{R}^n$ is finite. Observe that one may write $\text{cone}(E)$ in the form:

$$\text{cone}(E) = \{x \in \mathbb{R}^n : \exists \mu \in \mathbb{R}^{|E|} \text{ such that } x = \sum_{e \in E} \mu_e \cdot e \text{ and } \mu \geq 0_{|E|}\}$$

i.e., $\text{cone}(E)$ is the set obtained by projecting out the μ -variables (the last $|E|$ variables) from the polyhedral cone:

$$\left\{ \begin{pmatrix} x \\ \mu \end{pmatrix} \in \mathbb{R}^n \times \mathbb{R}^{|E|} : \sum_{e \in E} \mu_e \cdot e - x = 0_n \text{ and } \mu \geq 0_{|E|} \right\}.$$

It then follows from Theorem 1 that $\text{cone}(E)$ is a polyhedral cone.

Corollary 1 (*A finitely generated cone is a polyhedral cone*)

Finitely generated cones are polyhedral cones. ■